

Quant Foundations

This document is a comprehensive roadmap designed for individuals starting from zero (or near-zero) who want to build strong foundations for quantitative trading and quantitative research roles. The emphasis is deliberately placed on how quant firms actually evaluate candidates: problem-solving under uncertainty, probabilistic reasoning, coding for modelling, and applied projects rather than surface-level credentials.

This roadmap is not firm-specific coaching, but it is aligned with the expectations of top quantitative trading firms such as Jane Street, SIG, Optiver, IMC, Jump Trading, Citadel Securities, DRW, and similar proprietary trading firms.

PART I: WHAT “QUANT FOUNDATIONS” ACTUALLY MEANS

Quant foundations are not about knowing advanced mathematics in isolation. They are about building a mental toolkit that allows you to:

- Reason clearly about randomness**
- Model uncertain systems**
- Make decisions with incomplete information**
- Communicate structured thinking under pressure**

Most candidates fail not because they lack intelligence, but because their preparation is mis-ordered. They study advanced topics before mastering the core thinking patterns tested in interviews.

PART II: MATHEMATICAL FOUNDATIONS (INTERVIEW-FOCUSED)

1. Probability Fundamentals

Focus areas:

- Sample spaces and events
- Conditional probability
- Bayes' theorem
- Independence vs dependence
- Law of total probability
- Expectation, variance, covariance

You must be able to explain these concepts verbally, not just compute them.

Key interview skill:

Translate ambiguous word problems into clean probability models.

Example interview-style thinking:

“What assumptions am I making?”

“What happens if I condition on this event?”

“What symmetry can I exploit?”

2. Random Variables and Distributions

Core distributions to master:

- Bernoulli

- **Binomial**
- **Geometric**
- **Poisson**
- **Uniform**
- **Normal**
- **Exponential**

For each distribution:

- **Understand the story that generates it**
- **Know expectation and variance intuitively**
- **Know when it appears naturally in problems**

Interview emphasis:

Why is this distribution appropriate here?

3. Expectation as a Tool

Expectation is the most important object in quant interviews.

You should be comfortable with:

- **Linearity of expectation**
- **Expected stopping times**
- **Expected value vs risk**
- **Expected value under conditioning**

This appears constantly in puzzles, games, and trading-style questions.

PART III: QUANT INTERVIEW QUESTIONS (THE MOST IMPORTANT LAYER)

This is the single highest-ROI component of preparation.

Quant interview questions are not about recall. They are designed to test:

- Problem decomposition**
- Logical structure**
- Comfort with uncertainty**
- Communication clarity**

1. How to Approach Any Quant Question

A disciplined structure:

- 1. Restate the problem clearly**
- 2. Identify unknowns and assumptions**
- 3. Define the sample space**
- 4. Solve a simplified version**
- 5. Generalise**
- 6. Sense-check the result**

Interviewers care more about steps 1–4 than the final number.

2. Core Question Categories

You should practice heavily across:

- Coin and dice probability**

- Card problems
- Expected value games
- Optimal stopping problems
- Logical puzzles
- Market intuition questions

Depth beats breadth. Seeing variations of the same structure is more valuable than memorising answers.

3. Time Pressure Training

You must train yourself to:

- Speak while thinking
- Avoid silent computation
- Correct yourself out loud

Strong candidates narrate their reasoning clearly even when wrong initially.

PART IV: PYTHON AS A MODELLING TOOL

Python is not tested for syntax elegance. It is tested as a thinking amplifier.

1. Core Python Skills

- **Data structures (lists, dicts, sets)**
- **Loops and vectorisation**
- **Functions and modular code**
- **Basic object-oriented concepts**
- **Numerical libraries (NumPy, pandas)**

2. Python for Quant Thinking

Use Python to:

- **Simulate random processes**
- **Verify closed-form results**
- **Stress-test assumptions**
- **Visualise distributions**

If your code only reproduces known formulas, it is underused.

PART V: HIGH-IMPACT PYTHON PROJECTS (INTERVIEW-RELEVANT)

Projects should answer questions, not show tools.

1. Market Microstructure Simulator

Build a simplified limit order book.

Study:

- Spread behaviour
- Order imbalance
- Impact of random order flow
- Latency assumptions

Interview value:

Shows understanding of how markets actually work.

2. Volatility Surface Construction

- Compute implied volatility
- Build volatility smiles and surfaces
- Study skew dynamics across maturities

Interview value:

Signals derivatives literacy and numerical care.

3. Strategy Backtesting Engine

- Clean data handling
- Transaction costs
- Slippage
- Regime analysis

Interview value:

Demonstrates realism and discipline.

4. Monte Carlo Pricing Engine

- Simulate paths
- Compare analytical vs simulated results
- Study convergence

Interview value:

Links probability, coding, and finance.

5. Optimal Stopping Simulation

- Secretary problem
- Trading entry/exit decisions
- Risk of ruin analysis

Interview value:

Directly maps to interview questions.

PART VI: TOP QUANT FIRMS — WHAT THEY ACTUALLY LOOK FOR

Across firms, the signals are consistent:

- Clear probabilistic reasoning
- Comfort with abstraction
- Strong communication
- Evidence of independent thinking

They do not expect:

- Prior trading experience
- Finance degrees
- Perfect solutions

They do expect:

- Calm under pressure
- Structured reasoning
- Intellectual honesty

PART VII: CV AND INTERVIEW ALIGNMENT

Your CV should:

- **Highlight thinking, not tools**
- **Quantify outcomes**
- **Emphasise independent work**

In interviews:

- **Admit uncertainty**
- **Make assumptions explicit**
- **Guide the interviewer through your reasoning**

PART VIII: FINAL ADVICE

Quant preparation compounds slowly, then suddenly.

Most people quit just before the curve turns upward.

Focus on:

- Depth over breadth**
- Questions over content**
- Thinking over memorisation**